

ATAVISHAALA

FARM MARKET

# Strategic Roadmap & Execution Plan

Zero-Budget Agricultural Commerce & Crop Advisory Platform  
Phase 1 MVP · 4-Week Sprint · Manager's Strategic Reference

## TIMELINE

**4 Weeks**

20 working days · 4 sprints

## TEAM SIZE

**4 Developers**

MOB×1 · BE×2 · FE×1

## ARCHITECTURE

**Modular Monolith**

Node.js 20 · React Native · Postgres

## RECURRING INFRA COST

**₹0 / month**

8 free tiers, 0 paid services

## UNAVOIDABLE SPEND

**₹1,100 / year**

domain + transactional SMS

## RELEASE GATE

**14 Capabilities**

all must pass, no partial credit

This document is the **manager's half** of the Atavishaala delivery plan. It carries the milestone tracker, the sync cadence, the risk register, the cost audit and the release gate. Task-level instructions live in the companion volume, **Atavishaala — Team Execution Handbook**.

## MANAGER'S STRATEGIC REFERENCE

Cost audit · milestone tracker · sync framework · risk register · release gate

### § 1.0 Executive Summary

Atavishaala is a mobile-first agricultural commerce app with a consultation service attached. Farmers buy seeds, fertilizers, pesticides and equipment from verified companies, and — the part that actually differentiates us — they photograph a sick crop, describe the problem, and an agronomist calls them back within two hours. The storefront is commodity engineering. The Crop Advisor is the product.

That distinction drives every decision in this plan. Where the marketplace can be simplified, it is simplified: no public web storefront in Phase 1, no live GPS map, no supplier self-serve portal, no recommendation model. Where the Advisor is concerned, the media pipeline and the two-hour SLA engine get built in Week 2 and hardened for three weeks, because an SLA we miss is worse than an SLA we never promised.

#### What the ₹0 constraint actually buys — and costs

Every service in the stack sits on a permanent free tier, not a trial. The genuine monthly infrastructure cost is zero and stays zero to roughly 2,000 monthly active farmers and 1,000 advisory tickets. Two line items are not free and no amount of architecture will make them free: an **.in domain** (about ₹800/year) and **transactional SMS** in India, which requires TRAI DLT registration and is billed per message. We reduce the second by making **WhatsApp the primary OTP channel** — Meta's Cloud API gives 1,000 free service conversations a month, and WhatsApp is where our farmers already are. SMS becomes the fallback, not the default.

##### THE HONEST RISK

4 developers · 4 weeks · ~40 mobile screens, 8 backend modules and 2 web consoles. That is 16 developer-weeks against roughly 22 weeks of comfortable work. \$1.6 pre-agrees what gets cut so that **scope absorbs the slip, not quality**.

##### THE CORRECTION

The v2.0 handbook assumed Railway's free tier. Railway retired it — there is no standing free allowance any more. Phase 1 hosts the API on **Google Cloud Run** (always-free, scale-to-zero) with an Oracle Cloud always-free ARM box as the escape hatch.

One structural consequence of scale-to-zero hosting is worth flagging at management level, because it changes how we talk about reliability: **the API server is not always running**. It wakes on request in roughly a second and sleeps when idle. Nothing may therefore live inside the server process — no in-memory timers, no node-cron. All scheduled work is triggered externally by GitHub Actions against authenticated internal endpoints. This is a benefit, not a workaround: the jobs become independently observable and individually re-runnable.

### § 1.1 The ₹0 Claim, Audited

Every layer of the platform, the service chosen, and what we actually pay. Review this sheet before any budget conversation — it is the defensible version of "we run for free".

| LAYER                         | SERVICE (PHASE 1)              | FREE ALLOWANCE                          | MONTHLY COST | WHAT TRIGGERS SPEND                 |
|-------------------------------|--------------------------------|-----------------------------------------|--------------|-------------------------------------|
| <b>Mobile app</b>             | React Native + Expo SDK 53     | Unlimited local builds; EAS free queue  | ₹0           | Play Store listing: ₹2,100 one-time |
| <b>API hosting</b>            | Google Cloud Run (asia-south1) | 2M req · 360k GB-s · 180k vCPU-s / mo   | ₹0           | >2M requests/mo — far beyond MVP    |
| <b>Database</b>               | Neon Postgres (serverless)     | 0.5 GB storage · 190 compute-hr/mo      | ₹0           | 0.5 GB breached → \$19/mo           |
| <b>Media (images + video)</b> | Cloudinary                     | 25 credits/mo ≈ 25 GB storage+bandwidth | ₹0           | Advisor media retention not purged  |

| LAYER                         | SERVICE (PHASE 1)                        | FREE ALLOWANCE                      | MONTHLY COST    | WHAT TRIGGERS SPEND                             |
|-------------------------------|------------------------------------------|-------------------------------------|-----------------|-------------------------------------------------|
| <b>Product demo video</b>     | YouTube unlisted, embedded               | Unlimited, zero egress on us        | ₹0              | Never                                           |
| <b>Admin + Expert console</b> | Cloudflare Pages (Vite SPA)              | Unlimited bandwidth · 500 builds/mo | ₹0              | Never at our scale                              |
| <b>OTP — primary</b>          | WhatsApp Cloud API (Meta)                | 1,000 service conversations/mo      | ₹0              | >1,000 verifications/mo                         |
| <b>OTP — fallback</b>         | 2Factor.in / Fast2SMS                    | Trial credits only                  | variable        | <b>DLT ₹0 + ~₹0.15/SMS. Not free.</b>           |
| <b>Push notifications</b>     | Firebase Cloud Messaging                 | Unlimited                           | ₹0              | Never                                           |
| <b>Transactional email</b>    | Brevo                                    | 300 emails/day                      | ₹0              | >300/day                                        |
| <b>Scheduled jobs</b>         | GitHub Actions cron                      | 2,000 min/mo private repo           | ₹0              | Job runtime creep past 1,500 min                |
| <b>Queue / cache</b>          | Postgres (SKIP LOCKED + tables)          | Included in DB                      | ₹0              | Nothing — Redis deliberately absent             |
| <b>Payments</b>               | Razorpay Standard                        | No platform fee                     | <b>2% + GST</b> | <b>Per transaction. Unavoidable, COGS.</b>      |
| <b>Error monitoring</b>       | Sentry Developer                         | 5,000 errors/mo · 1 user seat       | ₹0              | Noisy release                                   |
| <b>Product analytics</b>      | PostHog Cloud                            | 1M events/mo                        | ₹0              | >1M events/mo                                   |
| <b>Uptime monitoring</b>      | UptimeRobot                              | 50 monitors · 5-min interval        | ₹0              | Never                                           |
| <b>DNS · SSL · WAF · CDN</b>  | Cloudflare Free                          | Unlimited                           | ₹0              | Never                                           |
| <b>Domain name</b>            | atavishaala.in (registrar)               | None                                | ~₹67            | ~₹800/yr. <b>Unavoidable.</b>                   |
| <b>Source control · CI</b>    | GitHub Free (private)                    | Unlimited private repos             | ₹0              | Never                                           |
| <b>TOTAL RECURRING</b>        | <b>18 services · 16 permanently free</b> | —                                   | <b>₹67/mo</b>   | <b>+ SMS fallback usage + 2% gateway on GMV</b> |

Read the bold rows as the answer to "so what does it really cost?" — **₹800 a year for the domain, ₹2,100 once for Play Store, per-message SMS only when WhatsApp fails, and 2% of GMV to Razorpay.** Everything else is genuinely ₹0 and stays that way while we are small.

## § 1.2 Visual Weekly Milestone Tracker

Every module mapped against the four sprints. A filled cell is active work with a named deliverable; an empty cell means the module is not this week's focus and should not be started. Open this table at every Monday kick-off and confirm reality matches it.

| MODULE / MILESTONE                           | DEV   | WEEK 1<br>Foundation       | WEEK 2<br>Catalog & Advisor | WEEK 3<br>Commerce Core | WEEK 4<br>Trust & Release  |
|----------------------------------------------|-------|----------------------------|-----------------------------|-------------------------|----------------------------|
| <b>Monorepo, CI/CD, environments</b>         | BE-1  | ■ Init + pipeline green    |                             |                         |                            |
| <b>Database schema (35 tables)</b>           | BE-1  | ■ Full schema + indexes    | ■ Catalog additions         |                         |                            |
| <b>Auth — OTP (WhatsApp + SMS)</b>           | BE-1  | ■ Send / verify / throttle |                             |                         |                            |
| <b>Auth — JWT, refresh rotation</b>          | BE-1  | ■ RS256 + rotation         |                             |                         |                            |
| <b>Profile, language, addresses</b>          | BE-1  | ■ 4-language profile       |                             |                         |                            |
| <b>Catalog: suppliers, categories</b>        | BE-1  |                            | ■ CRUD + ordering           |                         |                            |
| <b>Products, variants, pack sizes</b>        | BE-1  |                            | ■ CRUD + stock              |                         |                            |
| <b>Media pipeline (Cloudinary)</b>           | BE-1  |                            | ■ Signed direct upload      |                         |                            |
| <b>Search + filters (pg_trgm GIN)</b>        | BE-1  |                            | ■ GIN index + facets        |                         |                            |
| <b>Home feed, banners, wishlist</b>          | BE-1  |                            | ■ Feed assembly API         |                         |                            |
| <b>Crop Advisor — intake + media</b>         | BE-1  |                            | ■ Submit + 5img/3vid        |                         |                            |
| <b>Crop Advisor — SLA + assignment</b>       | BE-1  |                            |                             | ■ 2-hr engine live      | ■ Escalation polish        |
| <b>Notification transport (FCM + WA)</b>     | BE-1  |                            |                             | ■ Push on order events  | ■ All 9 templates          |
| <b>Cart, coupons, checkout pricing</b>       | BE-2  |                            |                             | ■ Pricing engine        |                            |
| <b>Order FSM + inventory reservation</b>     | BE-2  |                            |                             | ■ 6-state FSM + locks   |                            |
| <b>Razorpay + COD + webhooks</b>             | BE-2  |                            |                             | ■ Orders + idempotency  |                            |
| <b>Reconciliation + job recovery</b>         | BE-2  |                            |                             | ■ 10-min sweep          | ■ Load-tested              |
| <b>Wallet ledger + top-up</b>                | BE-2  |                            |                             | ■ Append-only ledger    |                            |
| <b>Refer &amp; Earn (₹50 / ₹100)</b>         | BE-2  |                            |                             |                         | ■ Full flow + fraud guards |
| <b>Reviews + post-delivery prompt</b>        | BE-2  |                            |                             |                         | ■ Write + moderate         |
| <b>Cancellations, returns + photos</b>       | BE-2  |                            |                             | ■ Cancel only           | ■ Returns + refund         |
| <b>Support tickets, FAQ, Write-to-us</b>     | BE-2  |                            |                             |                         | ■ Help centre live         |
| <b>Mobile: onboarding + OTP + profile</b>    | MOB-1 | ■ 7 screens                |                             |                         |                            |
| <b>Mobile: home, category, PDP, search</b>   | MOB-1 |                            | ■ 11 screens                |                         |                            |
| <b>Mobile: Crop Advisor (3 screens)</b>      | MOB-1 |                            | ■ Upload + submit           | ■ History + detail      |                            |
| <b>Mobile: cart, payment, orders, track</b>  | MOB-1 |                            |                             | ■ 9 screens             |                            |
| <b>Mobile: profile hub, wallet, settings</b> | MOB-1 |                            |                             |                         | ■ 12 screens               |
| <b>Admin: shell, auth, design system</b>     | FE-1  | ■ Login + TOTP             |                             |                         |                            |

| MODULE / MILESTONE              | DEV                                       | WEEK 1<br>Foundation                 | WEEK 2<br>Catalog & Advisor       | WEEK 3<br>Commerce Core | WEEK 4<br>Trust & Release |
|---------------------------------|-------------------------------------------|--------------------------------------|-----------------------------------|-------------------------|---------------------------|
| Admin: catalog + CSV importer   | FE-1                                      |                                      | ■ Bulk import 500 SKU             |                         |                           |
| Expert console (Advisor queue)  | FE-1                                      |                                      |                                   | ■ Queue + call log      |                           |
| Admin: ops order board          | FE-1                                      |                                      |                                   | ■ Status advance        | ■ Returns + refunds       |
| Admin: coupons, wallet, KPIs    | FE-1                                      |                                      |                                   |                         | ■ Full ops surface        |
| E2E test pass + bug bash        | All 4                                     |                                      |                                   |                         | ■ Zero P0/P1              |
| Production deploy + APK         | BE-1                                      |                                      |                                   |                         | ■ Release gate            |
| WEEK 1<br>Foundation & Identity | WEEK 2<br>Catalog, Media & Advisor Intake | WEEK 3<br>Commerce Core & SLA Engine | WEEK 4<br>Trust, Growth & Release |                         |                           |

## § 1.3 Weekly Sync Framework

Three fixed syncs a week. With only four developers, ceremony is expensive — so each sync has one job, a written output, and a named escalation trigger. Anything that does not fit these three slots goes in the async channel, not a new meeting.

| SYNC                              | CADENCE             | AGENDA — WHAT THE PM CHECKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | REQUIRED OUTPUT                                                                                                                                                                                                                                                                                                                      | ESCALATION TRIGGER                                                                                                                                                              |
|-----------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MONDAY<br/>Kick-off</b>        | Mon 09:00<br>45 min | <ol style="list-style-type: none"> <li>1. Replay Friday's demo — did it actually pass the gate?</li> <li>2. Read this week's row in the §1.2 tracker aloud; confirm each dev's 3 deliverables</li> <li>3. Freeze API contract changes for the week (log any into the OpenAPI diff)</li> <li>4. Confirm last week's blockers are closed, not carried</li> <li>5. Read the free-tier dashboards: Neon storage, Cloudinary credits, GitHub Actions minutes</li> <li>6. Confirm nothing from the §1.6 cut list has quietly crept back in</li> </ol> | <ul style="list-style-type: none"> <li>✓ Each dev has written 3 deliverables in the board</li> <li>✓ OpenAPI diff reviewed and merged, then frozen</li> <li>✓ Blocker list is empty or escalated with an owner</li> <li>✓ Free-tier usage recorded in the running log</li> <li>✓ Notes in the shared doc within 1 hour</li> </ul>    | A dev without 3 clear deliverables → PM assigns before the meeting ends. Contract dispute → freeze both sides and resolve in the room.                                          |
| <b>WEDNESDAY<br/>Blocker sync</b> | Wed 18:00<br>30 min | <ol style="list-style-type: none"> <li>1. Round-robin: DONE / IN PROGRESS / BLOCKED — one line each, no demos</li> <li>2. Name the tasks that will not make Friday's gate</li> <li>3. Surface FE↔BE and MOB↔BE integration gaps: are mocks replaced by real endpoints yet?</li> <li>4. Review the §1.4 risk matrix — has any probability moved?</li> <li>5. Check Sentry and the dev Cloud Run logs for new error classes</li> <li>6. Decide any scope reduction <b>now</b>, on Wednesday, not on Thursday night</li> </ol>                     | <ul style="list-style-type: none"> <li>✓ Every blocker has an assignee, not just a note</li> <li>✓ Friday-demo risk stated as a percentage</li> <li>✓ Scope cuts taken from §1.6 in order, and recorded</li> <li>✓ Pairing arranged for anything stuck 2 days</li> <li>✓ Board reflects true state before anyone logs off</li> </ul> | Two devs blocked on the same dependency → PM intervenes same day. Anything in the money path (payments, wallet ledger, order FSM) slipping → escalate to tech lead immediately. |
| <b>FRIDAY<br/>Demo gate</b>       | Fri 16:00<br>60 min | <ol style="list-style-type: none"> <li>1. Demo runs on a real Android device and the deployed API — never localhost, never a mock</li> <li>2. Walk the week's gate criteria from §1.8 one by one; each is pass or fail, nothing partial</li> <li>3. Confirm the week's migrations are applied to the shared database</li> <li>4. Record the free-tier usage delta for the week</li> <li>5. Log next week's known unknowns as explicit questions</li> <li>6. Fifteen minutes of retro: one thing to keep, one to change</li> </ol>               | <ul style="list-style-type: none"> <li>✓ Gate signed off in writing by the PM, or explicitly failed</li> <li>✓ Failed items become Monday's first assignment</li> <li>✓ Demo recording saved for stakeholders</li> <li>✓ Usage delta appended to the trend log</li> <li>✓ Retro action assigned to a person</li> </ul>               | A failed gate two weeks running → invoke §1.6 and cut a full feature, do not extend the timeline. A gate demoed on localhost does not count as passed.                          |

### WHAT IS NOT A MEETING

Daily standups. With four developers, a written async check-in at 10:00 in the team channel is faster and leaves a searchable record.

### THE ONE RULE

The Friday demo runs against deployed infrastructure. Every serious integration failure in a 4-week sprint hides behind the words "it works on my machine".

## § 1.4 Risk Mitigation Matrix

Risks specific to a four-week cycle, a four-person team and free-tier infrastructure. Reviewed every Wednesday. Probability and impact are assessed for *this* sprint, not in general — re-score them weekly rather than treating the table as static.

| RISK                                                                              | PROB.  | IMPACT          | MITIGATION                                                                                                                                                                                                                                                                                    | OWNER & TRIGGER                                                                         |
|-----------------------------------------------------------------------------------|--------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Scope vs. capacity</b> — 16 dev-weeks of team against ~22 weeks of listed work | HIGH   | <b>CRITICAL</b> | \$1.6 cut list is pre-agreed and ordered <b>before</b> Week 1 starts, so cutting is execution rather than negotiation. PM cuts on Wednesday, never Thursday. No feature enters scope mid-sprint under any circumstance.                                                                       | PM owns. Trigger: any Friday gate fails, or Wednesday risk to demo exceeds 30%.         |
| <b>Cloudinary credits consumed by Advisor video</b>                               | HIGH   | <b>HIGH</b>     | Three hard limits enforced client-side: <b>15 s max per clip, 720p, 3 clips</b> , compressed on device before upload. Server rejects anything over 8 MB. Purge job deletes Advisor media 30 days after a ticket resolves — this is what keeps storage flat instead of monotonic.              | BE-1 owns. Alert at 18 of 25 credits. Trigger: reduce to 10 s clips, then 2 clips.      |
| <b>Advisor 2-hour SLA missed in production</b>                                    | MEDIUM | <b>CRITICAL</b> | SLA clock runs on <b>business hours only (08:00-20:00 IST)</b> so a 21:00 submission is not born already failing. Sweep job every 10 min warns the expert at T-30 min and escalates to ops at breach. The confirmation screen states the window in the farmer's own language.                 | BE-1 builds, ops owns at runtime. Trigger: any breach in the first week of beta.        |
| <b>Razorpay webhook lost or duplicated</b>                                        | MEDIUM | <b>CRITICAL</b> | Webhook HMAC verified, event_id unique-indexed so replays are no-ops, and a reconciliation sweep every 10 min queries Razorpay for orders stuck in PENDING_PAYMENT past 15 min. Client callbacks <b>never</b> mutate money state.                                                             | BE-2 owns. Verified by integration test in Week 3, not by inspection.                   |
| <b>Neon 0.5 GB storage exhausted</b>                                              | LOW    | <b>HIGH</b>     | Media never touches Postgres — only Cloudinary IDs. Archive cron trims order_events, resolved consultations and webhook payloads older than 90 days. Realistic MVP footprint is 40-60 MB.                                                                                                     | BE-1 monitors weekly. Trigger at 350 MB: enable archival; at 450 MB: \$19/mo plan.      |
| <b>SMS OTP cost and DLT approval delay</b>                                        | MEDIUM | <b>MEDIUM</b>   | WhatsApp Cloud API is the primary channel, SMS the fallback — so a slow DLT approval does not block the sprint. Non-production environments use fixed OTP 123456. DLT template registration starts on Day 1 of Week 1 because approval takes 3-7 working days.                                | BE-1 files DLT Day 1. Trigger: not approved by end of Week 2 → ship WhatsApp-only beta. |
| RISK                                                                              | PROB.  | IMPACT          | MITIGATION                                                                                                                                                                                                                                                                                    | OWNER & TRIGGER                                                                         |
| <b>MOB-1 is a single point of failure</b> — one developer owns ~40 screens        | HIGH   | <b>CRITICAL</b> | FE-1 shares the React component conventions and can absorb mobile work from Week 3. Screens are built against mock fixtures so mobile never waits on backend. Every screen is merged the day it is built, never batched. MOB-1 documents navigation and state in the repo, not in their head. | PM owns. Trigger: MOB-1 absent 2+ days, or Week 2 screen count under 8 by Wednesday.    |

| RISK                                                                             | PROB.  | IMPACT | MITIGATION                                                                                                                                                                                                                                                                                                                                    | OWNER & TRIGGER                                                                                    |
|----------------------------------------------------------------------------------|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Scale-to-zero breaks in-process schedulers</b>                                | MEDIUM | HIGH   | Banned at review: node-cron, setInterval, in-memory sessions or queues. All scheduled work is a GitHub Actions cron calling POST /internal/jobs/:name with a shared secret. Validated in Week 1 so the constraint is discovered on day three, not day twenty.                                                                                 | BE-1 enforces in code review. Trigger: any PR containing a timer is rejected outright.             |
| <b>FE↔BE↔MOB integration bottleneck</b>                                          | HIGH   | HIGH   | OpenAPI spec committed and frozen by <b>end of Day 3, Week 1</b> . Both clients code against generated types and mock fixtures. Contract changes are a pull request against the spec, reviewed like code, and only on a Monday.                                                                                                               | BE-1 holds the spec; PM enforces the freeze. Trigger: any client blocked >4 hrs on a contract.     |
| <b>Multi-language content, not code</b> — 4 languages of copy nobody has written | HIGH   | MEDIUM | Phase 1 translates <b>UI strings and category names only</b> ; product names stay in English with brand names as printed on the pack, which is how farmers recognise them anyway. Ops supplies Hindi, Telugu and Kannada strings by Week 3 — this is a staffing task, not an engineering one, and it is the item most likely to be forgotten. | PM owns delivery of the strings file. Trigger: not delivered by Week 3 Monday → English-only beta. |
| <b>Referral fraud on Refer &amp; Earn</b>                                        | MEDIUM | MEDIUM | Referrer credit pays only after the referee's order is DELIVERED <b>and</b> the 7-day return window closes. Self-referral blocked on phone and device ID. Referral coupon valid on first order only. Monthly per-referrer credit cap is a config value, not a code change.                                                                    | BE-2 owns. Trigger: any referrer over 10 credits in a month is frozen for manual review.           |
| <b>Expo build failure in the release week</b>                                    | MEDIUM | MEDIUM | Expo SDK pinned; no dependency upgrades after Week 1 for any reason. A signed release APK is produced in <b>Week 2</b> , not Week 4, so the build path is proven with three weeks to spare. MOB-1 tests on a real mid-range Android device daily, not only in an emulator.                                                                    | MOB-1 owns. Trigger: no successful release APK by end of Week 2 → PM escalates.                    |
| <b>Cloud Run cold start hurts perceived speed</b>                                | MEDIUM | LOW    | Container kept lean so cold start stays near 1 s. Home feed served from a Cloudflare-cached response with a 5-minute TTL, so the first screen after app open does not wait on a wake-up. Health ping every 5 min from UptimeRobot keeps the instance warm during business hours at no cost.                                                   | BE-1 owns. Trigger: p95 first-request latency above 2.5 s in the Week 4 load test.                 |

§ 1.5 Free-Tier Capacity & Alert Thresholds

The failure mode of a zero-budget build is not a bill — it is a service silently refusing requests mid-sprint. Each row below names the number to watch, the level at which we act, and the specific degradation we take rather than reaching for a credit card. Read these figures out at every Monday kick-off.

| SERVICE            | HARD FREE LIMIT                                  | EXPECTED AT MVP                    | ACT AT     | PLANNED DEGRADATION                                                           |
|--------------------|--------------------------------------------------|------------------------------------|------------|-------------------------------------------------------------------------------|
| Neon Postgres      | 0.5 GB storage · 190 compute-hr/mo               | 40-60 MB · ~90 compute-hr          | 350 MB     | Archive order_events + webhook payloads >90 days, then \$19/mo                |
| Cloudinary         | 25 credits/mo (storage + bandwidth + transforms) | 8-12 credits at 1,000 tickets      | 18 credits | Drop clips to 10 s, then 2 clips; move product images behind Cloudflare cache |
| Cloud Run          | 2M req · 360k GB-s · 180k vCPU-s /mo             | ~150k req · well under compute cap | 1.2M req   | Cache home feed and catalog at Cloudflare edge (removes ~60% of traffic)      |
| WhatsApp Cloud API | 1,000 service conversations/mo                   | 600-800 at 2,000 MAU               | 800        | Route overflow OTP to SMS fallback; extend OTP validity to cut resends        |
| GitHub Actions     | 2,000 min/mo (private repo)                      | ~450 min at 10 deploys/wk          | 1,500 min  | Cache node_modules; skip CI on docs-only commits; widen cron intervals        |
| Cloudflare Pages   | Unlimited bandwidth · 500 builds/mo              | ~60 builds/mo                      | 400 builds | Build admin console on merge only, not on every push                          |
| Brevo email        | 300 emails/day                                   | ~40/day in beta                    | 220/day    | Batch order digests; make email optional where push already covers the event  |
| Sentry             | 5,000 errors/mo · 1 seat                         | ~400/mo                            | 3,500/mo   | Filter known client noise; sample non-critical breadcrumbs                    |
| PostHog            | 1M events/mo                                     | ~120k/mo                           | 700k/mo    | Drop low-value autocapture; keep funnel events                                |
| Expo EAS           | Free queue, limited concurrency                  | ~8 builds in the sprint            | —          | Build locally with eas build --local; EAS only for the release APK            |

WHO WATCHES THIS

BE-1 records every number in a single running spreadsheet each Friday. One person, one sheet, one habit — dashboards nobody owns get checked once and then never again.

THE ESCAPE HATCH

If two or more services breach their act-at line in the same week, the whole stack moves to one **Oracle Cloud always-free ARM instance** (4 cores, 24 GB) running Postgres, MinIO and the API under Docker Compose. Cost stays ₹0; ops burden rises. Budget two days.

## § 1.6 Pre-Agreed Scope Cut List

This is the most important table in the document. Four developers cannot comfortably build everything listed in Phase 1 in four weeks. Rather than discover that in Week 3 and argue about it, we agree the order of sacrifice now, while nobody is under pressure. **The PM cuts strictly from the top of this list, one item at a time, on a Wednesday.** Nothing below the line is touched until everything above it is gone.

| # | WHAT WE CUT                                      | WHAT THE FARMER STILL GETS                                                                                                                                                  | TIME RECOVERED | REVERSIBLE?                   |
|---|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------|
| 1 | <b>Rule-based "Recommended for you" feed</b>     | The section shows top-rated products in the categories they last bought from — a single SQL query instead of a scoring pass.                                                | 2 days         | Yes, Phase 2                  |
| 2 | <b>Product review photos</b>                     | Star rating plus text review. Photo upload on reviews is the least used surface in the app and it costs us Cloudinary credits.                                              | 1.5 days       | Yes, Phase 2                  |
| 3 | <b>In-app help centre article browser</b>        | Help & Support opens a WhatsApp deep link to the ops number and a short static FAQ screen.                                                                                  | 2 days         | Yes, Phase 2                  |
| 4 | <b>Wishlist</b>                                  | Products can still be added to cart and left there. Wishlist is a nice-to-have that touches four screens.                                                                   | 2 days         | Yes, Phase 2                  |
| 5 | <b>Coupon engine beyond the referral coupon</b>  | Referral discount works. Ops-issued promo codes wait — no marketing campaign is running in beta anyway.                                                                     | 3 days         | Yes, Phase 2                  |
| 6 | <b>Post-delivery returns with photo evidence</b> | Cancellation before dispatch works fully. Post-delivery returns are handled by ops over WhatsApp with a manual wallet credit.                                               | 4 days         | Yes, but ops carries the load |
| 7 | <b>Wallet top-up (Add Money)</b>                 | The wallet still receives refunds and referral credit and can still pay for an order. Farmers just cannot load their own money into it — which almost none will do in beta. | 3 days         | Yes, Phase 2                  |
| 8 | <b>Multi-language beyond English + Hindi</b>     | Telugu and Kannada ship in the first post-launch release. The language picker stays; two options are greyed with "coming soon".                                             | 3 days         | Yes — it is a content task    |

| BELOW THE LINE — NOT CUTTABLE UNDER ANY CIRCUMSTANCE                | WHY                                                                                         |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Crop Advisor intake, media upload and the 2-hour SLA engine</b>  | It is the product. Everything else is a storefront anyone could build.                      |
| <b>Order state machine correctness and inventory reservation</b>    | Overselling stock or losing an order state destroys trust we cannot rebuy.                  |
| <b>Payment idempotency, webhook verification and reconciliation</b> | A double-charged farmer is an unrecoverable customer and a legal exposure.                  |
| <b>Wallet ledger integrity — append-only, balance derived</b>       | A wallet whose balance cannot be explained from its ledger is unauditible.                  |
| <b>OTP throttling, JWT rotation and rate limiting</b>               | The cheapest attack surface on the internet is an unthrottled OTP endpoint.                 |
| <b>Delivery confirmation geofence (200 m)</b>                       | It is the only control preventing an agent from marking a parcel delivered from their home. |

## § 1.7 MVP Definition of Done

Fourteen capabilities define "production-ready" for Atavishaala Phase 1. Each has criteria that are observable rather than described — a thing you can run and watch either pass or fail. All fourteen must be signed off before the Week 4 release gate. **There is no partial credit:** a capability at 90% is not done, it is a Monday task.

| CAPABILITY                               | ACCEPTANCE CRITERIA — OBSERVABLE, NOT DESCRIBED                                                                                                                                                                                                                                                                                                                                                          | VERIFIED BY                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>1 • Authentication</b>                | POST /auth/otp/send returns 200 and a WhatsApp message arrives on a real handset. POST /auth/otp/verify returns an RS256 access token plus refresh token. OTP is stored bcrypt-hashed — confirmed by reading the row in psql. Sixth request within an hour returns 429. Refresh-token reuse outside the 30 s grace window revokes the whole family. Tokens live in Expo SecureStore, never AsyncStorage. | BE-1 + MOB-1, integration test on device |
| <b>2 • Onboarding &amp; language</b>     | New number walks Get Started → mobile → OTP → personal info → 3 slides → home. Returning number skips straight to home. Language choice persists across a cold app restart and switches UI strings and category names immediately.                                                                                                                                                                       | MOB-1 manual walkthrough, 2 devices      |
| <b>3 • Catalog &amp; discovery</b>       | 500 SKUs imported by CSV without error. Home feed returns categories, banners, companies and recommendations in a single request under 400 ms. Search for a partial brand name returns results and EXPLAIN ANALYZE confirms the trigram GIN index is used, not a sequential scan. Category and price filters compose correctly.                                                                          | BE-1 + MOB-1, EXPLAIN output attached    |
| <b>4 • Product detail</b>                | Image gallery swipes, YouTube demo video plays, pack-size variants switch price and stock, rating breakdown matches the reviews table, wishlist and share both work, out-of-stock variant disables Add to Cart rather than failing at checkout.                                                                                                                                                          | MOB-1 + BE-1 manual test                 |
| <b>5 • Crop Advisor intake</b>           | 5 photos and 3 videos upload direct to Cloudinary — confirmed in the network inspector that bytes never traverse our API. A 30-second raw clip is compressed on device to under 2 MB and rejected if over 15 s. Submission returns a ticket number and states the 2-hour window in the farmer's chosen language.                                                                                         | BE-1 + MOB-1 on a real 4G connection     |
| <b>6 • Advisor SLA engine</b>            | sla_due_at computed on business hours only — a 21:00 submission is due 10:00 next day, verified by unit test. Assignment picks an expert matching language and district with the lowest load. Sweep job warns at T-30 min and escalates at breach. Expert console shows a live countdown, and the call log writes back to the farmer's advisory history.                                                 | BE-1 + FE-1, unit tests + console demo   |
| <b>7 • Cart &amp; pricing</b>            | Cart survives app restart. Adding the same variant twice increments quantity rather than duplicating the line. Referral coupon applies exactly ₹50 and is rejected on a second order. Server recomputes every total from scratch and ignores any client-supplied price — proven by tampering with the request.                                                                                           | BE-2 + MOB-1, tampered-request test      |
| <b>8 • Order state machine</b>           | CONFIRMED → PACKED → DISPATCHED → OUT_FOR_DELIVERY → DELIVERED runs end to end. assertTransition() rejects any skip with 422. Ten concurrent orders for one unit of stock: exactly one succeeds, nine get 409. Every transition writes an order_events row, and the mobile timeline is rendered from those rows rather than from a duplicated status field.                                              | BE-2, concurrent load test script        |
| CAPABILITY                               | ACCEPTANCE CRITERIA — OBSERVABLE, NOT DESCRIBED                                                                                                                                                                                                                                                                                                                                                          | VERIFIED BY                              |
| <b>9 • Payments &amp; reconciliation</b> | Razorpay test-mode payment completes and the order moves to CONFIRMED only after the webhook, never on the client callback. Replaying the same webhook returns 200 with no state change. A deliberately dropped webhook is recovered by the 10-minute reconciliation sweep. COD orders skip payment and are marked PENDING_COD. Every amount in the database is an integer in paise.                     | BE-2, Razorpay dashboard + replay test   |
| <b>10 • Wallet integrity</b>             | Refund on cancellation credits the wallet and writes one ledger row. SUM(credits) – SUM(debits) equals wallet_accounts.balance_paise for every user — asserted by a test over 200 synthetic transactions. Paying with wallet debits atomically and cannot go negative under concurrent requests. Ledger rows are never updated or deleted.                                                               | BE-2, reconciliation assertion test      |
| <b>11 • Delivery confirmation</b>        | Agent confirming delivery outside 200 m of the address is rejected; inside 200 m succeeds and stores the measured distance. Farmer receives a push within seconds, and the review prompt arrives on the delivered items.                                                                                                                                                                                 | BE-2 + MOB-1, GPS spoof test             |

| CAPABILITY                                       | ACCEPTANCE CRITERIA — OBSERVABLE, NOT DESCRIBED                                                                                                                                                                                                                                                                                                                                     | VERIFIED BY                           |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>12 · Refer &amp; Earn</b>                     | Referral code is unique per user and shareable. Referee gets ₹50 off their first order only. Referrer's ₹100 lands in the wallet only after the referee's order is DELIVERED and the 7-day return window has closed. Self-referral by the same phone is blocked.                                                                                                                    | BE-2, end-to-end with 2 test accounts |
| <b>13 · Notifications</b>                        | FCM push received on a real device for: order confirmed, dispatched, out for delivery, delivered, review prompt, refund credited, advisory assigned. WhatsApp template delivers OTP and the advisory callback confirmation. A stale FCM token is marked inactive on first send failure rather than retried forever.                                                                 | BE-1 + MOB-1 on a real device         |
| <b>14 · Security, deploy &amp; observability</b> | CI runs lint → typecheck → test → build → deploy green on main. Cloud Run health endpoint returns 200. Cloudflare SSL active on api. and admin.. Helmet headers present. No secret in the repository — verified by a history scan, not by eye. Logs redact phone, address and payment identifiers. Sentry and UptimeRobot both live. Signed release APK installs on a clean device. | BE-1 + PM sign-off                    |

## § 1.8 Weekly Sprint Goals & Friday Demo Gates

One sentence per week describing what becomes true, and the demo that proves it. If the demo cannot be performed on the deployed environment on a real device, the week did not pass — regardless of how much code was merged.

| WEEK          | THEME                                      | SPRINT GOAL — WHAT BECOMES TRUE                                                                                                                                                                                                                                                                                                                                                                                                                  | FRIDAY DEMO GATE                                                                                                                                                                                                                                                                    |
|---------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEEK 1</b> | <b>Foundation &amp; Identity</b>           | The scaffolding is trustworthy. Every developer runs the API locally against the shared Neon database. All 35 tables exist with their indexes. A farmer can enter a phone number, receive an OTP over WhatsApp, land on a profile screen, choose a language and a district, and save an address. The OpenAPI contract is frozen. CI deploys to Cloud Run on merge.                                                                               | Live from a real handset: phone → WhatsApp OTP → profile → language → saved address. \dt in psql shows 35 tables. CI pipeline green on main. A GitHub Actions cron successfully calls an internal job endpoint.                                                                     |
| <b>WEEK 2</b> | <b>Catalog, Media &amp; Advisor Intake</b> | The app has something to sell and the Advisor accepts work. Ops imports 500 SKUs by CSV. A farmer browses categories and companies, searches by brand, opens a product, switches pack sizes, and watches the demo video. Separately, they photograph a sick crop, attach three short clips, and submit a ticket that appears in the database with media attached.                                                                                | Home → category → search → product detail → variant switch, all on real API data. Admin imports a CSV of 500 SKUs. Advisor submission with 5 photos and 3 videos completes over 4G, with the network inspector proving bytes bypassed our API. Signed release APK installs.         |
| <b>WEEK 3</b> | <b>Commerce Core &amp; SLA Engine</b>      | Money moves correctly and experts have a queue. A farmer fills a cart, applies a referral coupon, pays with Razorpay in test mode or chooses COD, and receives an order. Ops advances it through five states and the farmer's timeline follows. Stock cannot be oversold, a payment cannot be lost or counted twice, and the wallet balance always equals its ledger. The expert console shows tickets ordered by SLA deadline with a countdown. | Cart → coupon → Razorpay test payment → CONFIRMED, then ops advances to DELIVERED with the mobile timeline updating. Replayed webhook is a no-op. Ten concurrent orders for one unit: one succeeds, nine 409. Wallet ledger reconciles to the paisa. Expert console countdown live. |
| <b>WEEK 4</b> | <b>Trust, Growth &amp; Release</b>         | The loop closes and the system ships. Delivered orders prompt reviews that appear on the product page. Cancellations and returns issue wallet refunds. Referrals pay out on qualification. The profile hub, wallet history, saved addresses, advisory history, help centre and account settings are all live. Security is hardened, monitoring is on, and the APK is signed.                                                                     | <b>RELEASE GATE:</b> production URL live behind Cloudflare SSL, signed APK installed on a clean device, full farmer journey completed end to end by someone who did not build it, load test passed, Sentry clean, all fourteen §1.7 capabilities signed off in writing.             |

## § 1.9 Phase 2 Backlog & When to Start Spending

Phase 1 exists to answer one question: do farmers submit advisory tickets and buy the products the expert recommends? Everything below waits for evidence. The right-hand column names the signal that justifies the spend — spending ahead of the signal is how zero-budget discipline is usually lost.

| PHASE 2 ITEM                                   | WHY IT WAITS                                                                                                                                                  | COST WHEN WE DO IT                    | SIGNAL THAT UNLOCKS IT                                      |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------|
| <b>Public web storefront with SEO</b>          | Mobile is where farmers are; web SEO is a 3-month payback, not a 4-week one.                                                                                  | ₹0 — Cloudflare Pages                 | Organic search interest, or a distributor asking for a link |
| <b>Live GPS tracking on a map</b>              | The five-state timeline answers "where is my order" for 95% of queries at a fraction of the build cost.                                                       | ₹0 infra, ~2 weeks build              | Support volume dominated by "where is it right now"         |
| <b>Supplier self-serve portal</b>              | Six to ten suppliers can be onboarded by ops over WhatsApp. A portal is worth building at forty.                                                              | ₹0 infra, ~3 weeks build              | More than 20 active suppliers                               |
| <b>Server-side video transcoding &amp; ABR</b> | Genuinely cannot be done free. Client-side compression at 720p is adequate for diagnosing leaf damage.                                                        | ~\$25-40/mo Cloudinary or Mux         | Experts reporting they cannot diagnose from current quality |
| <b>ML crop-disease pre-classification</b>      | Needs a labelled dataset we do not have yet. Phase 1 is how we collect it — every ticket is a labelled example.                                               | ₹0 to start (fine-tune on free Colab) | 3,000+ resolved tickets with expert labels                  |
| <b>Wallet cash-out / bank withdrawal</b>       | <b>Regulatory, not technical.</b> A closed-loop store credit needs no licence; letting money out is a prepaid payment instrument and needs RBI authorisation. | Legal + compliance cost               | Never, unless the business model requires it                |
| <b>Split-tender payment (wallet + gateway)</b> | Two payment legs on one order roughly doubles the reconciliation surface. Phase 1 wallet is all-or-nothing.                                                   | ₹0 infra, ~1 week build               | Wallet balances routinely smaller than order values         |
| <b>Automated gateway refunds</b>               | Refund-to-wallet is instant, costs nothing and keeps money inside the ecosystem.                                                                              | ₹0 — Razorpay refund API              | Farmers asking for money back to their bank                 |
| <b>In-app voice calling for advisory</b>       | Farmers already use phone and WhatsApp. VoIP adds cost and a failure mode on rural bandwidth.                                                                 | ~₹0.50/min via a CPaaS                | Experts unable to reach farmers by phone                    |
| <b>Elasticsearch / typo-tolerant search</b>    | Postgres trigram search handles a few thousand SKUs comfortably.                                                                                              | ₹0 self-hosted (Meilisearch)          | More than 5,000 SKUs, or measurable search abandonment      |

**Build the Advisor properly. Everything else is a storefront.**

Atavishaala · Farm Market — Phase 1 MVP · Manager's Strategic Reference